

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fifth Semester of B. Tech IT/CE Examination****March-April 2018****IT303/303.01/303.02 Advanced Database Management System****Date: 03.04.2018, Tuesday****Time: 01.30 p.m. To 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 Do as directed. [07]**
- a. **Define the following terms(Any three) [03]**
1. Speed Up
 2. Scale Up
 3. Dense index
 4. Sparse index
- b. Indices whose search key specifies an order different from the sequential order of the file are called _____ indices. [01]
- a) Nonclustered
 - b) Secondary
 - c) All of the mentioned
 - d) None of the mentioned
- c. A transaction manager is which of the following? [01]
- a) Maintains a log of transactions
 - b) Maintains before and after database images
 - c) Maintains appropriate concurrency control
 - d) All of the above
- d. Differentiate: Primary index v/s Secondary index [02]
- Q-2.a A B+ tree of order 4 is built for (1, 4, 7, 10, 17, 21, 31, 25, 19, 20, 28, and 42). [07]**
- What is the maximum number of node splitting operations that may take place after constructing final B+ tree?
 - What is final B+ tree after deleting 42 and 28?
- Q-2.b List various techniques for Data Fragmentation and explain Vertical Fragmentation in detail with example. [04]**
- Q-2.c Explain Multilevel index with example and appropriate figure. [03]**

OR

- Q-2.b Explain in brief with neat sketch Shared memory and Shared disk parallel database architecture. [04]**
- Q-2.c Differentiate Homogenous v/s Heterogeneous Distributed Database System. [03]**
- Q-3.a Discuss two-phase commit (2PC) protocol with example and for each possible failure, explain how 2PC ensures transaction atomicity. [06]**

Q-3.b Answer any TWO questions.**[08]**

- Define data server. Also list various issues and explain any explain page shipping v/s item shipping in detail.
- List various background processes with respect of transaction server. Explain lock manager and log writer process in detail.
- Consider the total bucket size of 10 with one record per bucket (i.e. $k \bmod 10$). Insert the following no 3, 7, 33, 45, 15, 67. Solve the problem of overflow bucket using linear and quadratic probing.

SECTION – II**Q - 4 Do as directed.****[07]****a. Define the following terms(Any four)****[04]**

- Soft Deadline
- Intra-query Parallelism
- Partition Skew
- Attribute value Skew
- Inter-query Parallelism

b. Which of the following is not the advantage of parallel processing?**[01]**

- a) Scale up b) Autonomy c) Speed up d) Availability

c. What is the purpose of view? Explain with example.**[02]**

Q-5.a Explain and compare partitioning techniques of parallel database using scan, point and range queries with the help of example. **[06]**

Q-5.b Explain symmetric and asymmetric fragment and replication join with neat sketch in detail. **[04]**

Q-5.c List types of parallel sort in Intra-operation parallelism and explain any one of them with neat sketch. **[04]**

OR

Q-5.b Explain Transaction Workflows with example. **[04]**

Q-5.c List TP-Monitor Architectures and justify how “Many server, single router model” overcomes dis-advantage of “Single process model”. **[04]**

Q-6.a Draw detail structure of TP monitor and explain in detail roles of various managers. **[06]**

Q-6.b Answer any TWO questions.**[08]**

- Define skew and how virtual processing technique handle skew.
- Use RSA public key encryption algorithm with $p=11$, $q=5$, $e=7$ for plaintext “WORLD” and generate public and private key pair. Encrypt plain text using generated key pair.
- Explain compensation transaction with example. Does it solve cascading rollback problem?
